

STUDENT RESULT MANAGEMENT SYSTEM

- Python + SQLite Project
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PROJECT OVERVIEW

- A menu-driven student result management system built using Python.
- SQLite is used to store student records permanently.
- The system supports adding, viewing, updating and deleting student data.
- It also calculates total marks, percentage, grade and pass/fail result.

TECHNOLOGIES USED

- Python — application logic and menu-driven interface
- SQLite3 — lightweight database for storing records
- SQL — CREATE, INSERT, SELECT, UPDATE and DELETE operations
- Functions — separate modules for each operation

MAIN FEATURES

- Add Student — stores roll number, name and marks of 3 subjects
- View Students — displays all saved student records
- Update Marks — changes marks using the roll number
- Delete Student — removes a student record
- Calculate Result — computes total, percentage, grade and result

RESULT CALCULATION LOGIC

- $\text{Total} = \text{Subject 1} + \text{Subject 2} + \text{Subject 3}$
- $\text{Percentage} = \text{Total} \div 3$
- PASS requires at least 33 marks in every subject
- Grade: A+ (90+), A (75–89), B (60–74), C (33–59), F (<33)
- Database changes are saved using `conn.commit()`

CONCLUSION

- The project demonstrates practical use of Python with a database.
- It provides basic CRUD operations for student records.
- The modular functions make the program easy to understand and extend.
- Future improvements: GUI, input validation, more subjects and report export.
- Thank You